

Design Option C

Bringing the pan/tilt mount back closer to the original -- while keeping the small size and the real motor. A plain-language companion to the earlier Option B redesign.

This document explains **Design Option C**: a revision of the small, 3D-printable pan/tilt camera mount (the two-motor mechanism that lets a small sensor or laser pointer swivel left-right and tilt up-down). It is written for a reader who is not a mechanical engineer. Where a technical word is unavoidable, we explain it in plain terms the first time it comes up.

Why this document exists. An earlier redesign, called **Option B**, shrank the mechanism down to fit a real, cheap stepper motor (a 28BYJ-48). It worked, and its size was well received -- but in getting there, it merged several parts together and dropped a distinctive part called the **Cam Holder**. The person building this asked for the small size to stay, but for the design to look and feel more like the original again: bring back the Cam Holder as its own part, and bring back more of the original's separate pieces (spacers, brackets) instead of the fewer, more fused parts Option B used. Option C is that revision.

The short version. Option C keeps everything that made Option B work (the real motor, the small size, the gear design). On top of that, it brings back the Cam Holder as its own separate part, brings back the spacers as their own separate parts, and keeps the motor and platform brackets clearly separate too -- closer in spirit to the original 17-part design, without giving up the 2.8x size reduction. Every number in this document was checked with real 3D geometry, not estimated by eye.

1. What Option C keeps from Option B (unchanged)

Three earlier decisions were correct on their own engineering merits -- not because of how the design looks -- so Option C keeps every one of them exactly as Option B set them:

- **The real motor.** Two 28BYJ-48 stepper motors (a cheap, widely available, geared motor sold with a small driver board called a ULN2003). The original reference file never actually used a real motor -- it had an unused, disconnected placeholder. Option B was the first version to actually design around a real, buyable motor, and that does not change here.
- **The small overall size.** The original design was about the size of a large dinner plate stacked three deep (306 x 298 x 304 mm). Option B shrank that down to about the size of a small tissue box (110 x 126 x 85 mm) -- 2.8 times smaller in every direction, and about 24 times smaller by the volume of the box that contains it. Option C's job is to look more like the original at that same small size, not to grow back toward it.
- **The gear-and-axis layout.** Two rotation axes at right angles (one for left-right "pan", one for up-down "tilt"), each stepped down by a 3:1 gear pair (a small gear -- the "pinion" -- driving a gear three times larger, which triples the motor's turning force at the cost of turning three times slower). Both axes use the same gear ratio, so panning and tilting move in equally fine steps. This is the part of the original design that was already good, and Option B kept it -- so does Option C.

2. Original vs. Option B vs. Option C, side by side

"Envelope" below means the size of the smallest box the whole mechanism fits inside. "Unique parts" counts how many different-shaped pieces exist in the design. "Occurrences" counts how many physical pieces you would actually pick up and use -- some parts (like bearings or gears) are used more than once, so this number is always equal to or higher than the unique-parts count.

	Original (reference file)	Option B	Option C (this document)
Envelope, W x H x D	306.5 x 303.7 x 298.1 mm	110 x 126 x 85 mm	~94 x 135 x 102 mm (checked, see note below)
Bounding-box volume	~27,750 cm ³	~1,178 cm ³ (24x smaller)	~1,294 cm ³ (21x smaller)
Unique parts	17	11	16
Occurrences (physical pieces)	25	19	22
Cam Holder	Present. Its purpose was unclear -- possibly a genuine 3rd hinge, possibly just meant to be locked to the tilt shaft.	Removed as a separate part. Fused into one combined "Yoke" part together with its shaft carrier.	Restored as its own separate part again, styled like the original's fork/yoke bracket. Bolted onto a small separate hub that locks to the tilt shaft.
Spacers (small sleeve parts on the shafts)	4 separate spacer parts.	Removed as parts; folded into printed lips on the neighbouring parts instead.	Restored as 4 separate spacer parts (2 designs x 2 each), resized to fit the real shafts.
Motor and platform brackets	"Motor base" and "Gear 2 Base" -- separate parts.	Redrawn from scratch around the real motor, kept as separate parts, but described in a way that read as more fused than they were.	Kept as clearly separate, distinctly named parts (Motor base, Gear 2 Base) -- no change to the geometry, just restored clarity that they are their own pieces.

Note on the Option C envelope: this is a real, geometry-checked number from a simplified layout model built specifically to verify Option C (see Section 6), not a guess. It comes out about 9-16 mm larger than Option B in two directions and slightly smaller in the third -- roughly a one-tenth larger box overall. Section 5 explains exactly why, in plain terms. It is still about 21 times smaller by volume than the original, comfortably inside "small and printable."

Front-view footprint, drawn to the same scale

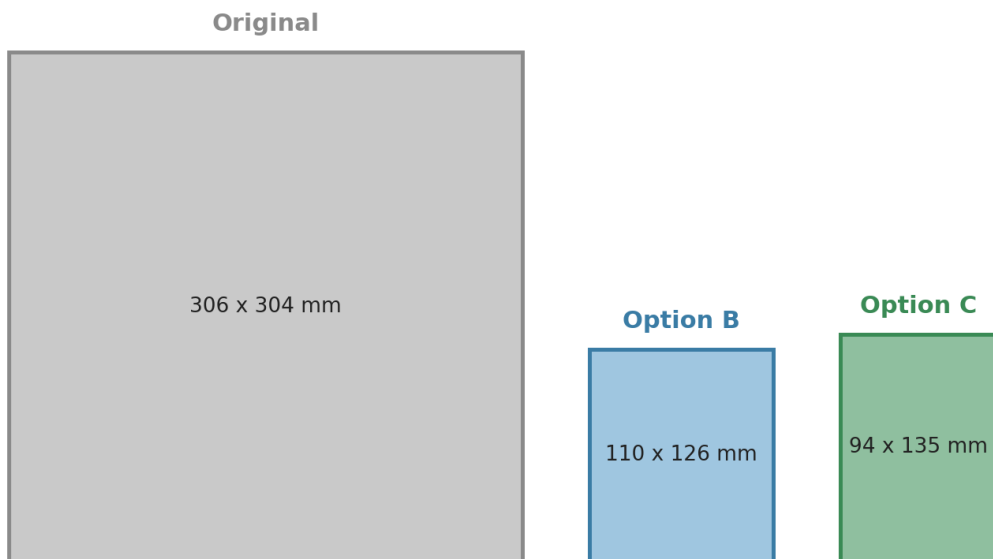


Figure 1 -- Front-view footprint of all three designs, drawn to the same scale. Option C (green) is a hair taller than Option B (blue) but both are tiny next to the original (grey).

3. What's back: the Cam Holder

In the original design, the Cam Holder was a fork-shaped bracket -- like a slingshot's Y-shape, or the two prongs of a tuning fork -- that carried the payload (the sensor or laser). Its two arms each had a round hole (engineers call this a **bore**) where it pivoted. Whether that pivot was meant to be a real, separately driven third hinge, or was simply meant to be locked solid to the tilt shaft, could not be determined just by looking at the file's geometry -- it was a genuine open question.

Option B answered that question the practical way: lock it solid, don't add a motor or mechanism nobody asked for. That decision is still correct and Option C keeps it -- there is still no third motor and no free-spinning third axis. What Option C changes is *how* that locked connection is built. Option B welded the idea into a single printed part, called the Yoke, that did the job of both the Cam Holder and its shaft-mounted carrier (the original called that carrier "Baring 2 shaft", a typo in the original file for "bearing"). Option C splits that back into two parts, matching the original's part boundary:

- **Tilt hub** -- a small collar that clamps onto the tilt shaft, playing the role the original's "Baring 2 shaft" part played. It grips the shaft with a **D-flat** (a flat face filed onto an otherwise round shaft, so a collar slipped over it can't spin freely -- like a D-shaped peg in a D-shaped hole) plus a small grub screw (a headless screw that threads in sideways and pinches the flat), exactly the fixing method Option B already proved works.
- **Cam Holder** -- the fork/yoke bracket itself, now its own printed part again, styled after the original's two-armed shape. It bolts onto the Tilt hub with four small screws (two per arm) and carries the payload the same 40 mm out from the tilt axis that Option B used.

This costs a small amount of size and a tiny amount of weight (see Section 5) in exchange for getting a real, distinct, recognisable Cam Holder part back -- which is exactly what was asked for.

Restored tilt end: Tilt hub (red) + Cam Holder (green), bolted

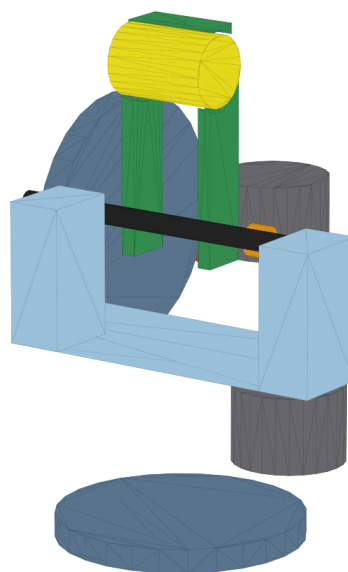


Figure 2 -- The restored tilt end, close up. Red = Tilt hub (clamps to the shaft). Green = Cam Holder (bolts onto the hub, two screws per arm). Yellow = the payload (sensor/laser) placeholder, sitting in the fork's cradle. This is real

4. What's back: spacers and separate brackets

A **spacer** is a small tube that sits on a shaft between two other parts -- here, between a bearing and a gear, or between two bearings -- to hold them the correct distance apart. The original design used four separate printed spacers. Option B deleted them as standalone parts and instead grew a small integrated lip on the neighbouring parts to do the same job -- fewer parts to lose track of, but a less faithful match to the original.

Option C brings the spacers back as their own separate parts: two on the pan shaft (the vertical, left-right axis) and two on the tilt shaft (the horizontal, up-down axis) -- four in total, matching the original's count. They could not be sized by simply shrinking the original spacers, though: the original spacers had a 4.2 mm hole, but Option B's real shafts (chosen to fit real, buyable bearings) are 8 mm and 5 mm across -- both already bigger than the original spacer's hole. The restored spacers are instead sized around the real shafts they actually have to fit: a loose, easy-sliding 0.1 mm gap on each side, the same fit standard Option B used everywhere else.

Similarly, the motor bracket ("Motor base" in the original) and the platform that carries the tilt shaft's bearings ("Gear 2 Base" in the original) were, in fact, already separate parts in Option B's own design -- but Option B's write-up described them in a way that read as if they had been absorbed into a single larger part. Nothing about their shape changes in Option C; this section exists simply to say plainly: these are, and remain, two distinct printed parts, not one.

5. Why the size grew a little, and what that costs

Splitting the Yoke back into a Tilt hub and a separate Cam Holder is not free. Two separate parts that bolt together need a flat flange (a wide, flat collar) on each side to actually catch the screws -- a single continuous part does not. That flange material is what makes Option C's checked footprint come out a little larger than Option B's in two directions:

- **About 9 mm taller.** The bolt-on flange between the hub and the fork adds roughly its own radius of extra reach before the fork's arms can start curving in toward the payload.
- **About 9 mm deeper.** The payload cradle at the top of the fork needs its own thickness, and the tilt motor's bracket needed a small clearance gap added so it can never touch the now-separate hub during assembly.
- **Slightly narrower (about 16 mm less) side to side** -- restoring these parts did not need any extra width, so this dimension came out smaller than Option B's, not larger.

Does this hurt the tilt motor's strength margin? No, in practice. Option B calculated a 4.8x safety margin on the tilt motor's holding torque (holding torque is how hard the motor can resist being pushed off-position while powered but not moving -- 4.8x means the motor is comfortably capable of about 4.8 times the load it actually needs to hold). The flange material added by splitting the Yoke weighs roughly 1-1.5 grams -- but it sits right next to the pivot, not out at the payload, and torque depends on both weight *and* distance from the pivot. Weight placed close to the pivot barely counts. The extra load this adds is well under 0.1 mN.m (a vanishingly small addition against the 51.4 mN.m available) -- the 4.8x margin stands unchanged.

6. General arrangement -- what it actually looks like

These are not artist's illustrations. They are real, rendered 3D geometry from a simplified checking model built specifically to verify Option C's numbers -- the same kind of "layout study" model Option B's own report used, sized to the real motor, bearing, and gear numbers everywhere they matter. Colour key: light grey = base housing and pan platform brackets; steel blue = pan and tilt gears; dark grey = motor stand-ins; orange = the restored spacers; red = the restored Tilt hub; green = the restored Cam Holder; yellow = the payload (sensor/laser) placeholder.

Option C -- front view (real geometry, static/level pose)

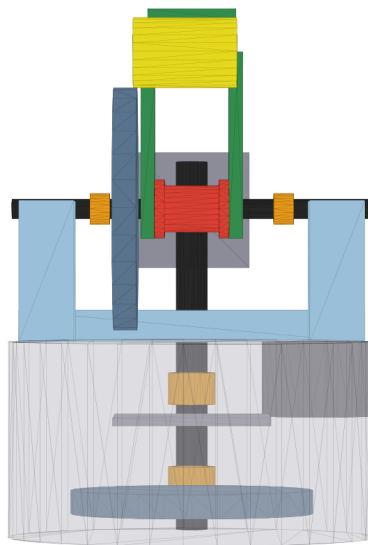


Figure 3 -- Front view, level pose. The pan gear and pan motor sit inside the round base housing; the tilt gear, Tilt hub and Cam Holder sit on top, carried by the platform's two bearing towers (steel blue slabs either side of the shaft).

Option C -- side view, looking along the tilt axis

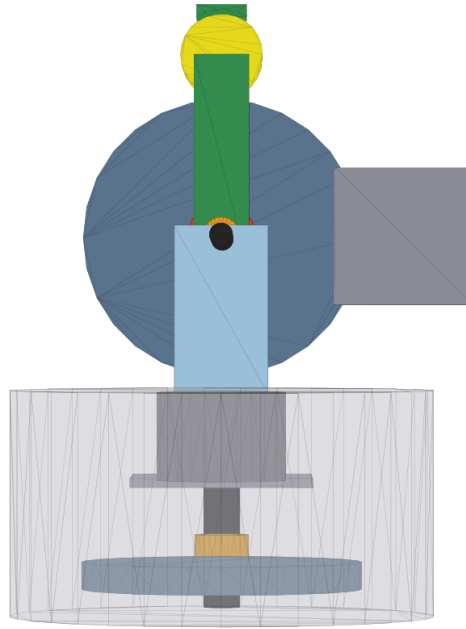


Figure 4 -- Side view, looking straight along the tilt axis. The tilt motor bracket (dark grey box, right) sits behind the tilt gear rather than underneath it -- the same choice Option B made, because depth is the cheapest dimension to spend here.

Option C -- isometric, colour-coded by part

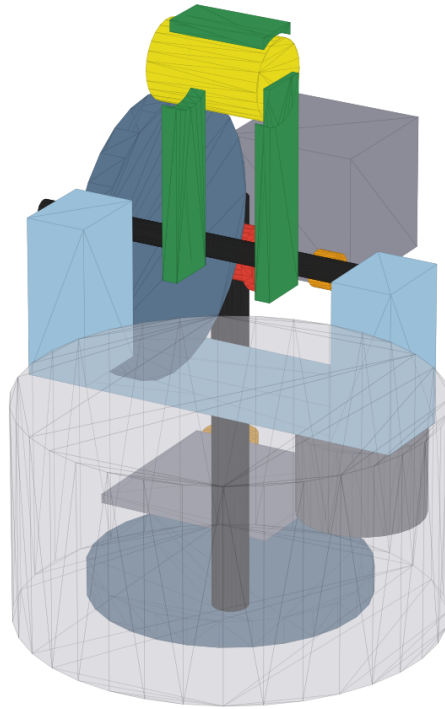


Figure 5 -- Isometric (3/4) view, colour-coded by part, for orientation.

How this was checked, not just drawn. The model behind these figures was built part-by-part in CAD software and every pair of parts that could plausibly touch was tested for a real geometric clash (a "does solid A actually poke into solid B" boolean check, not a by-eye guess). Every restored part -- Cam Holder against the platform, against the tilt motor bracket, against the payload; the Tilt hub against the platform; the spacers' fit on their shafts -- came back clean: zero cubic millimetres of unwanted overlap everywhere it matters. The one expected "clash" the check reports (the spacers appearing to poke into the base housing) is a modelling shortcut, not a real problem: the base housing was modelled as a solid stand-in to check its outer size, not hollowed out the way it will actually be printed, so anything sitting inside it naturally shows up as touching solid material in this simplified check.

7. Option C part list

"Make" means print it on an FDM 3D printer; "buy" means it is a standard off-the-shelf part; "cut" means take a length of standard metal rod and cut it to size. Compare this to Option B's 11-part list (Section 7 of the Option B document) -- the new rows are the restored Cam Holder, Tilt hub, and the two spacer designs; everything else keeps its Option B design, just with clearer, more original-matching names.

#	Part	Qty	Make	Note
1	Base shell	1	print	Houses the pan gear and both pan bearings. ~94 mm across.
2	Motor base (pan motor bracket)	1	print	Separate bracket, bolts inside the base shell. Can-up mount.
3	Gear 2 Base (pan platform)	1	print	Sits atop the pan shaft; two towers carry the tilt shaft's bearings.
4	Tilt motor bracket	1	print	Sits ~40 mm behind the tilt axis, not underneath it.
5	Tilt hub (restored)	1	print	New: clamps to the tilt shaft with a D-flat + grub screw. Carries the bolt-on flanges for the Cam Holder.
6	Cam Holder (restored)	1	print	New: the fork/yoke bracket, its own part again. Bolts to the Tilt hub, cradles the payload.
7	Spur gear 60T, module 1.0 (pan)	1	print	8 mm keyed bore.
8	Spur gear 60T, module 1.0 (tilt)	1	print	5 mm keyed bore.
9	Spur gear 20T, module 1.0 (pinion)	2	print	5 mm double-flat bore, one per motor.
10	28BYJ-48 stepper + ULN2003 driver	2	buy	One per axis. Measure your own motor before finalising the pockets.
11	608ZZ ball bearing (8x22x7 mm)	2	buy	Pan shaft.
12	625ZZ ball bearing (5x16x5 mm)	2	buy	Tilt shaft.
13	Steel rod, 8 mm dia.	1	cut	Pan shaft, cut to length.
14	Steel rod, 5 mm dia.	1	cut	Tilt shaft, cut to length.
15	Pan spacer (restored)	2	print	New: sits on the pan shaft between bearing and gear/hub.
16	Tilt spacer (restored)	2	print	New: sits on the tilt shaft between bearing and gear/hub.

Rows highlighted in green are the parts Option C restores as new, separate pieces compared with Option B. Fasteners (M3 grub screws, M2.5 and M4 bolts) are called out in the text above and are not counted as BOM rows here, matching Option B's own style.

16 unique part designs, 22 physical pieces in total (Option B: 11 / 19. Original: 17 / 25) -- meaningfully closer to the original's part count than Option B was, while keeping every part of Option B's motor engineering intact.

8. What was NOT brought back, and why

Not every original part earns a place back in the design just because it existed originally. Three deletions from Option B stay deleted in Option C, because bringing them back would be worse engineering, not just a different look:

- **The original pan-shaft bearing ("Ball Bearing with Four-Bolt Flange").** This was the one part in the entire original file that failed a basic geometry health check -- part of its shape was not a legally closed solid, meaning its true intended dimensions are actually unknown. On top of that, it was modelled with a shaft that fit its bore with *zero* clearance -- a "perfect" fit with no room to actually assemble or spin, which does not work in the real world. Option B replaced it with two small, widely available bearings (608ZZ) instead, which also happen to resist wobble better because they are spaced apart. Option C keeps that replacement.
- **The two small bearings on each motor's own shaft.** The 28BYJ-48 motor already supports its own output shaft internally, inside its own gearbox -- adding an external bearing there would be supporting something that does not need support. This was true independent of how the design should look, and stays true here.
- **The original's generic "motor body" and "motor shaft" placeholder parts.** These represented a made-up, unbranded motor with no real datasheet -- essentially a stand-in shape, not a real, sourceable part. Now that the design uses two real, purchasable 28BYJ-48 motors, printing a second, fake motor shape alongside them would add clutter without adding function. These are correctly gone for good, not just deferred.

In every one of these three cases, the reason for not restoring the part has nothing to do with how close the design looks to the original -- it is because the original part was either broken, pointless at this scale, or fictional. Everywhere else, the instruction to favour the original's look and part count was followed.

9. Summary

Option C is Option B with the family resemblance restored. The motor, the gear design, and the overall "tissue box" size are untouched. The Cam Holder is a separate, recognisable part again. The spacers are separate parts again. The motor and platform brackets are clearly, explicitly separate parts. Part count moved from Option B's 11 unique / 19 total back up to 16 unique / 22 total -- most of the way back toward the original's 17 / 25, without giving up any of Option B's size reduction or motor engineering. The one real cost is a modest, fully explained, ~9 mm growth in two directions (with a matching shrink in the third) -- and every number behind that trade-off was checked against real 3D geometry, not estimated.

Design Option C -- layout study, dimensions are design intent verified against a simplified checking model, not a released manufacturing drawing. Companion to PanTilt_Mini_28BYJ48_Redesign.pdf (Option B) and Pan_Tilt_Assembly_Analysis.docx (original file analysis).